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The Plant Genome

ORIGINAL RESEARCH

Identification of genomic regions associated with the plasticity of carbon 13 ratio in soybean

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Abstract

Improving water use efficiency (WUE) for soybean [*Glycine max* (L.) Merr.] through selection for high carbon isotope (C13) ratio may increase drought tolerance, but increased WUE may limit growth in productive environments. An ideal genotype would be plastic for C13 ratio; that is, be able to alter C13 ratio in response to the environment. Our objective was to identify genomic regions associated with C13 ratio plasticity, C13 ratio stability, and overall C13 ratio in two panels of diverse Maturity Group IV soybean accessions. A second objective was to identify accessions that differed in their C13 ratio plasticity. Panel 1 (205 accessions) was evaluated in seven irrigated and four drought environments, and Panel 2 (373 accessions) was evaluated in four environments. Plasticity was quantified as the slope from regressing C13 ratio of individual genotypes against an environmental index calculated based on the mean within and across environments. The regression intercept was considered a measure of C13 ratio over all environments, and the root mean square error was considered a measure of stability. Combined over both panels, genome-wide association mapping (GWAM) identified 19 single nucleotide polymorphisms (SNPs) for plasticity, 39 SNPs for C13 ratio, and 16 SNPs for stability. Among these SNPs, 71 candidate genes had annotations associated with transpiration or water conservation and transport, root development, root hair elongation, and stomatal complex morphogenesis. The genomic regions associated with plasticity and stability identified in the current study will be a useful resource for implementing genomic selection for improving drought tolerance in soybean.

1 | INTRODUCTION

Soybean [*Glycine max* (L.) Merr.] is one of the most important legume crops in the world with grain that is a source of high-quality protein and vegetable oil. Several abiotic and biotic constraints limit soybean yield. Among the abiotic constraints, drought is the most important yield-limiting factor (Mohammadi et al., 2012; Pathan et al., 2007). Drought can cause yield losses in soybean of 80% or higher, depending upon the severity and stage when drought occurs (Dias et al., 2012; Oya et al., 2004), which has the potential to affect food security for the growing world population. Although crop management practices such as improved irrigation methods

Abbreviations: EI, environmental index; GWAM, genome-wide association mapping; G×E, genotype × environment interaction; LD, linkage disequilibrium; MG, maturity group; QTL, quantitative trait locus; RMSE, root mean square error; SNP, single nucleotide polymorphism; WUE, water use efficiency.

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(Jha et al., 2016), managing soils to increase water holding capacity (Ramakrishna et al., 2006), adjustment of planting dates (Prasad et al., 2017; Zhang et al., 2019), and selection of specific crop maturity responses (Bobojonov et al., 2013) have helped ameliorate drought effects, the most cost-effective method for alleviating the impact of drought on crop yield is through breeding (Vogel et al., 2021).

Efforts to improve soybean drought tolerance based strictly on yield have largely been unsuccessful (Arya et al., 2021; Wei et al., 2018; Yu et al., 2021). Identifying drought-tolerant genotypes based directly on yield is a challenge because of low heritability of yield under drought conditions and significant interactions between genotype and environment (Ceccarelli et al., 1991). An alternative strategy is to breed soybean genotypes for physiological traits that enhance drought tolerance (Sinclair et al., 2004). Several different morpho-physiological traits, such as carbon isotope ratio (Bai & Purcell, 2018a; Bazzer et al., 2020a, 2020b; Dhanapal et al., 2015a; Kaler et al., 2017b, 2018a), canopy wilting scoring (Abdel Haleem et al., 2012; Chamarthi et al., 2021; Charlson et al., 2009; Du et al., 2009; Hwang et al., 2015, 2016; Kaler et al., 2017a; Steketee et al., 2020; Ye et al., 2020), canopy temperature measurements (Bai & Purcell, 2018b; Bazzer & Purcell, 2020; Kaler et al., 2018b; Ludlow & Muchow, 1990; Pinto et al., 2010), and relative water content of leaves (Babu et al., 2003) represent opportunities for use as proxy selection methods for genetic enhancement of drought tolerance.

Among the physiological traits, water use efficiency (WUE) is a trait associated with drought tolerance (Blum, 2009; Sinclair, 2012; Vadez et al., 2014) whereby the quantity of dry matter produced per unit of water transpired by the crop is defined as WUE. Hence, WUE is associated with yield due to the positive relationship of seed yield with total crop mass (Chen et al., 2011; Passioura, 1996). Although high WUE can be used as a criterion to select drought-tolerant genotypes (Condon et al., 2002, 2004), screening for WUE on a large scale in the field is difficult and can be prohibitively expensive (Chen et al., 2011).

Farquhar & Richards (1984) proposed that carbon isotope discrimination ($\Delta^{13}C$, ‰) in C3 plants was due to differences in the diffusion between $^{13}CO_2$ and $^{12}CO_2$ (a, ~4.4‰), ribulose 1,5-bisphosphate (RuBP) carboxylase discrimination against the heavier $^{13}CO_2$ molecule (b, ~27‰), and the ratio of CO_2 inside (C_i) and outside of the leaf (C_a):

$$\Delta^{13}C = a + (b - a)\left(1 - C_i/C_a\right) \quad (1)$$

When considering instantaneous gas exchange in leaves, WUE can be expressed as:

$$\text{WUE} = \left[C_a\left(1 - C_i/C_a\right)\right] / \left[1.6\left(e_i - e_a\right)\right] \quad (2)$$

Core Ideas

- Increased C13 ratio may increase water use efficiency and drought tolerance but may limit growth.
- An ideal genotype would be plastic for C13 ratio (i.e., change C13 ratio in response to the environment).
- We found large differences among accessions for C13 plasticity, stability, and overall C13 ratio.
- Genome-wide association mapping identified seven loci for C13 ratio plasticity and eight for stability.
- Candidate genes from association mapping were linked to transpiration, water conservation, rooting, and stomates.

where e_i and e_a refer to the water vapor concentration inside the leaf and in the surrounding atmosphere, respectively, and 1.6 is a constant accounting for the differences in diffusivity between the heavier CO_2 molecule and H_2O. Hence, there is an inverse association of $\Delta^{13}C$ with WUE due to a common dependence upon C_i/C_a; that is, as C_i/C_a decreases, $\Delta^{13}C$ decreases and WUE increases. Isotopic fractionation may also be expressed as a ratio between ^{13}C relative to ^{12}C (C13 ratio, ‰), in which case there is a positive association between C13 ratio and WUE (Bazzer et al., 2020b). The relationship between $\Delta^{13}C$ or C13 ratio and WUE has been documented in numerous C3 crops including wheat (*Triticum aestivum* L.; Condon et al., 1990), bean (*Phaseolus vulgaris* L.; Polania et al., 2016; White, 1993) cowpea [*Vigna unguiculata* (L.) Walp.] (Ismail et al., 1994) and peanut (*Arachis hypogea* L.; Wright et al., 1994).

A high C13 ratio (i.e., high WUE) can result from a high photosynthetic rate that decreases the CO_2 concentration inside the leaf (C_i, Equation 1) to low concentrations. The greater WUE of C4 plants compared with C3 plants is due to a considerably lower C_i. Alternatively, high WUE may result from low stomatal conductance that decreases the flux of CO_2 into the leaf, resulting in a decreased C_i. Constitutive low stomatal conductance expectantly results in high WUE, which may be an advantage under drought conditions. However, low stomatal conductance under conditions of plentiful soil moisture will limit photosynthesis and may ultimately limit crop yield (Condon et al., 2004).

Plasticity refers to the ability of a genotype to express different phenotypes under different environmental conditions. Differences in plasticity among genotypes are the natural result of differing genotype × environment (G×E) interactions (Becker & Leon, 1988). Plasticity has been quantified using linear regression methods proposed by Finlay & Wilkinson

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(1963) and Eberhart & Russell (1966). Eberhart & Russell (1966) regressed individual genotype yield of maize (*Zea mays* L.) versus an environmental index (EI) for each environment by taking the mean of all genotypes at one environment minus the grand mean of all environments. From this analysis, the slope of the regression for a given genotype versus the EI quantifies the trait plasticity across environments whereby a greater slope indicates a greater plasticity. The intercept indicates the average trait value over all environments. The root mean square error (RMSE) measures the stability of the trait response over environments for each genotype and indicates how close the observed data points are to the regression line. A smaller RMSE indicates a greater stability than does a larger RMSE (Bernardo, 2002; Eberhart & Russell, 1966). An ideal genotype would be able to increase WUE in response to unfavorable environmental conditions but maintain high stomatal conductance (and relatively low WUE) under favorable conditions. That is, WUE and C13 ratio would ideally have plastic responses to the environment.

The availability of next-generation sequencing technologies, easily accessible genomic tools, and improved phenotyping methods has provided a path to understand G×E interactions at both the genomic level (Cuevas et al., 2017; Deshmukh et al., 2014), and at the quantitative trait locus (QTL) level. Genome-wide association mapping (GWAM) has emerged as a robust option to identify marker-trait associations among diverse germplasms. Genome-wide association mapping has identified QTLs associated with numerous traits in soybean (Chamarthi et al., 2021; Dhanapal et al., 2015a, 2015b; Kaler et al., 2017a, 2017b, 2018a, 2018b, 2018c; Steketee et al., 2020; Ye et al., 2020), and has also identified genomic loci associated with plasticity for various traits in rice (*Oryza sativa*; Kadam et al., 2017; Kikuchi et al., 2017), sunflower (*Helianthus annuus* L.; Mangin et al., 2017), maize (Gage et al., 2017; Kusmec et al., 2017), wheat (Lozada & Carter, 2020), and for yield in soybean (Gaston et al., 2021; Xavier et al., 2018) and wheat (Sehgal et al., 2020).

The present research evaluated the response of soybean C13 ratio to diverse environments in two panels consisting of 205 accessions, evaluated in 11 environments, and 373 accessions, evaluated in four environments. Our objectives were to: (a) determine genomic regions associated with C13 ratio plasticity (slope), stability (RMSE), and C13 ratio averaged over all environments (intercept), and (b) identify individual genotypes that differed in the response of C13 ratio to diverse environments.

2 | MATERIALS AND METHODS

The soybean genotypes used in the current study were divided into two panels. Panel 1 consisted of 205 Maturity Group (MG) IV accessions used for the evaluation of the C13 ratio in 11 environments that included seven irrigated and four drought environments (Chamarthi et al., 2021). Panel 2 consisted of 373 MG IV accessions used for the evaluation of the C13 ratio in four environments that included two irrigated environments and two rainfed environments (Dhanapal et al., 2015a; Kaler et al., 2017b).

2.1 | Panel 1

2.1.1 | Experimental material

Chamarthi et al. (2021) discussed in detail the selection of 205 MG IV soybean accessions used in Panel 1. In addition to 199 accessions, Panel 1 also included six checks: PI 416937 (slow wilting), PI 471938 (slow wilting), A5959 (fast wilting), 08705_16 (fast wilting breeding line), LG11-8169-007F (MG IV elite breeding line), and Lee non-nod (non-nodulating check, PI 573285). LG11-8169-007F and Lee NN were planted in each of 12 incomplete blocks in each environment using an augmented incomplete block design; all other genotypes were randomly associated to an incomplete block within each of the three replications. Of the 205 accessions in Panel 1, 99 accessions were chosen based on genotypic data from Panel 2, which consisted of 373 accessions used by Kaler et al. (2017b). Therefore, 99 accessions were included in both panels, and 100 accessions in Panel 1 were chosen from the USDA-GRIN collection (https://npgsweb.ars-grin.gov/) based on predicted extremes for canopy wilting (Kaler et al., 2017a), nitrogen (N_2) fixation (Dhanapal et al., 2015b), canopy temperature (Kaler et al., 2018b), and C13 ratio (Kaler et al., 2017b) from high breeding values calculated from previous association mapping studies.

2.1.2 | Locations and management

The experimental materials were planted at four locations during the 2018 and 2019 cropping seasons: (1) the Pine Tree Research Station (PT), AR (35.2547 N, −90.7965), (2) the Rohwer Research Station (RH), AR (33.8102 N, −91.2777), (3) the Bradford Research Center at Columbia (CO), MO (38.897 N, −92.2180), and (4) the Maricopa Agricultural Center, Univ. of Arizona at Maricopa (MC), AZ (33.0689 N, −111.9722).

At each location, phosphorus and potassium were applied as needed based on the soil test results, and pesticides were applied as needed. At the PT and the RH locations, each accession was planted in nine-row plots with a plot length of 4.57 m and with 19 cm between rows. At the CO location, each accession was planted in four-row plots with a plot length of 3.96 m and with 38 cm between rows. At the MC location, each accession was planted in three-row plots with a plot

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length of 4.87 m and with 19 cm between rows. The experiments had irrigated (IR) and drought (DR) environments that were planted in side-by-side blocks at each location. Irrigation was accomplished using drip irrigation at CO, flood irrigation at PT, and furrow irrigation at RH and MC. At the PT and the RH locations, both IR and DR treatments received irrigation before the V6 stage when the estimated soil moisture deficit exceeded 50 mm (Purcell et al., 2007); irrigation was terminated for the DR treatment at the V6 stage. Irrigation for the IR treatment at MC was applied every 7 d, and because of the high evaporative demand, irrigation for the DR treatment was applied every 14 d.

We did not include or evaluate 2018 DR treatment data from PT and RH, and 2019 DR treatment data from CO in the analysis due to frequent rainfall during the season that eliminated drought stress. There was no experiment conducted in 2019 at RH. Hereafter, we refer to the combination of site-year and treatment as an environment. Out of seven site-years, four site-years had IR and DR treatments and three site-years had only the IR treatment. These 11 environments were designated for 2018 as IR (CO18IR) and DR (CO18DR) at Columbia, MO; IR (MC18IR) and DR (MC18DR) at Maricopa, AZ; IR (PT18IR) at Pine Tree, AR; and IR (RH18IR) at Rohwer, AR. Likewise, designations were made for 2019 as, IR (CO19IR) at Columbia, MO; IR (MC19IR) and DR (MC19DR) at Maricopa, AZ; and IR (PT19IR) and DR (PT19DR) at Pine Tree, AR. Planting dates, sample collection dates, temperatures (min and max), total precipitation, number of irrigations, and cumulative potential evapotranspiration for 11 environments of Panel 1 and for four environments of Panel 2 are reported in Supplemental Table S1. To estimate the soil-moisture deficit for the DR treatment at each site year, we calculated the cumulative potential evapotranspiration rate between emergence and plant sampling (Purcell et al., 2007).

2.2 | Panel 2

2.2.1 | Experimental material

Dhanapal et al. (2015a) and Kaler et al. (2017b) described the 373 MG IV soybean accessions used in Panel 2. Briefly, 373 accessions were planted in two field locations using a randomized complete block design with two replications per location.

2.2.2 | Locations and management

The experimental material was planted at two locations during 2009 and 2010 at (1) the Bradford Research and Extension Center near Columbia (CO), MO (38°53'N, 92°12'W) and (2) the Rice Research Experiment Station near Stuttgart (ST), AR (34°30 N, 91°33' W). At both locations phosphorus and potassium were applied to meet soil test recommendations, and pesticides were applied as needed following recommended agronomic practices. At the CO location, each accession was planted in four-row plots with a plot length of 4.87 m and with 76 cm between rows. At the ST location, each accession was planted in a single row with a plot length of 6.1 m and 76 cm between rows. At the CO location, the experiment was rainfed, and at the ST location, the experiment was furrow irrigated as needed. The 2 yr and two locations were considered as four separate environments. These four environments were designated as CO09DR and CO10DR for the DR treatment at Columbia, MO; and ST09IR and ST10IR for the IR treatment at Stuttgart, AR.

2.2.3 | Sample collection and phenotyping

The aboveground portion of five plants were harvested between full bloom (R2) and beginning seed (R5) for Panel 1 and beginning bloom (R1) and full bloom (R2) for Panel 2 (Fehr et al., 1971) from each genotype in each replication at all locations. Plant samples were dried at 60 °C until a constant weight, and then ground using a multistep process (Dhanapal et al., 2015a) to obtain a finely powdered sample. Approximately 3 mg of the finely powdered sample was transferred to a tin capsule and sent to the University of California Davis for isotope analysis. Measurement of the absolute isotope composition is difficult; therefore, the C13 ratio was expressed relative to the international standard of the $^{13}C/^{12}C$ ratio Vienna PeeDee Belemnite (V-PDB). The website of the Stable Isotope facility (https://stableisotopefacility.ucdavis.edu/) provides more information and details.

2.3 | Statistical analysis

The PROC REG procedure of SAS version 9.4 was used to regress the mean C13 ratio of individual genotypes within an environment against the EI values to determine the slope, intercept, and RMSE for individual genotypes from Panel 1 and Panel 2. The EI was calculated for each environment as the mean C13 ratio for that environment minus the grand mean of all genotypes over all environments (Eberhart & Russell, 1966). After initial regression analysis, PROC REG was used to identify potential outliers based upon data points with studentized residuals >2.5. In total, 14 data points were removed out of a total of 2,234 data points from Panel 1. After the removal of outliers, the regression analysis was repeated to obtain slope, intercept, and RMSE values used for GWAM. No outliers were removed from Panel 2 due to fewer environments. Analysis for k-means clustering was performed using

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an initial range of 4 to 20, and k-means = 16 was found to be optimal via Cubic Cluster Criterion and applied to each of the two panel datasets. Clustering and scatterplots were generated using the JMP 16.0 software package (JMP 16, SAS Institute).

2.4 | Genotyping and linkage disequilibrium

We searched for the 205 genotypes from Panel 1 and for the 373 genotypes from Panel 2 individually in SoyBase (Glyma.w82.a2.v1, www.soybase.org) for publicly available data generated using the Illumina Infinium SoySNP50K iSelect single nucleotide polymorphism (SNP) Beadchip (Song et al., 2013, 2015). Out of the 205 genotypes from Panel 1, 201 genotypes had marker data, and all 373 genotypes comprising Panel 2 had marker data that were downloaded. The SNPs were subjected to quality control using TASSEL 5.0 (Bradbury et al., 2007) by removing the SNPs with minor allele frequency (MAF) ≤5%, SNPs with a missing rate higher than 10%, and monomorphic and heterozygous SNPs. After quality control, 34,035 SNPs for Panel 1 and 31,295 SNPs for Panel 2 were obtained. Missing data with ≤10% in the filtered SNPs were imputed using an LD-kNNi method (Money et al., 2015). Filtered and imputed SNPs were then used for GWAM to identify the significant SNPs for slope, intercept, and RMSE. The current study used the same linkage disequilibrium (LD) results as published by Kaler et al. (2020a).

2.5 | Genome-wide association mapping

We selected the FarmCPU model in the R package (Liu et al., 2016) as an appropriate model to control false positives and false negatives (Kaler & Purcell, 2019; Kaler et al., 2020b; Liu et al., 2016). Significant SNPs were identified using a threshold value of $-\text{Log10}\,P \geq 3.5$, which is equivalent to a P-value ≤ .0003. Similar threshold values with FarmCPU have been successfully used in previous studies (Chamarthi et al., 2021; Kaler & Purcell, 2021; Kaler et al., 2017a, 2017b, 2018b, 2018c, 2020a; Steketee et al., 2020). For each significant SNP, the allelic effect was calculated by taking the difference in mean C13 ratio between genotypes with the major allele and those with the minor allele. A positive sign in the allelic effect indicated that the major allele was associated with an increased slope, intercept, or RMSE from the regression analysis, and a negative sign in the allelic effect indicated that the minor allele was associated with an increase slope, intercept, or RMSE from the regression analysis.

To find coincident SNPs or overlapping loci identified in this study with loci reported in previous studies (Bazzer & Purcell, 2020; Bazzer et al., 2020b; Chamarthi et al., 2021; Kaler et al., 2017a, 2017b, 2018b; Steketee et al., 2020) and Panel 1 with Panel 2 and vice versa, we used Bedtools Intersect Intervals Tool (Quinlan & Hall, 2010) in Galaxy with an overlapping window of ± 175 kb. This window size was chosen because the average LD across all chromosomes decayed at an average of 175 kb in the euchromatic region (Kaler et al., 2020a).

2.6 | Candidate gene identification

To find the candidate genes linked to significant SNPs associated with the slope, intercept, and RMSE, we searched each SNP in the genome browser (www.soybase.org) using Glyma.Wm82.a2 version of soybean genome assembly (Schmutz et al., 2010) and examined a window size up to 100 kb. Genes within the window that were associated with drought and WUE were considered potential candidate genes. Candidate genes were then grouped based on Gene Ontology (GO) terms for biological process, cellular location, and molecular function. Based on GO biological function, genes were identified as having direct or indirect association with drought and its related responses.

3 | RESULTS

3.1 | Phenotypic descriptions

3.1.1 | Panel 1

We observed large differences in average maximum and minimum temperatures and total precipitation between emergence and the biomass sample collection dates among the 11 environments (Supplemental Table S1). The average maximum temperature was highest for MC18 and MC19 (37 ˚C) and lowest for CO18 and CO19 (29 ˚C), whereas the average minimum temperature was highest for MC18 (24 ˚C) and lowest for CO19 (12 ˚C). Total precipitation between emergence and the biomass sample collection was highest for CO19 (457 mm) and lowest for MC18 (15 mm). To quantify drought for each environment, we estimated the cumulative potential evapotranspiration (Purcell et al., 2007), which ranged from 414 mm (RH18) to 731 mm (MC19).

The wide range in temperatures, rainfall, and potential evapotranspiration resulted in a range of average C13 ratios among environments from −26.75‰ (CO18DR) to −29.21‰ (MC19IR) and EI values that ranged from 1.30 (CO18DR) to −1.16 (MC19IR) (Supplemental Table S2). For all four environments that had both IR and DR treatments, the average C13 ratio of the DR treatment was significantly ($P \leq .01$) greater than that of the IR treatment (Figure 1), which was expected since drought stress is known to increase WUE and C13 ratio

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TABLE 1 Descriptive statistics for the plasticity (slope), carbon isotope (C13) ratio (intercept), and stability (root mean square error, RMSE) values for regression of C13 ratio versus the environmental index for individual accessions in Panel 1, which consisted of 205 genotypes across 11 environments, and for individual accessions in Panel 2, which consisted of 373 accessions across four environments

Statistic	Plasticity	C13 ratio	Stability	r^2[a]	SE plasticity[b]	SE C13 ratio[b]	−Log10(p)[c]
		‰				‰	
Panel 1							
Mean	1.00	−28.05	0.29	.82	0.15	0.09	4.24
Median	1.00	−28.03	0.29	.86	0.14	0.09	4.28
Minimum	0.45	−28.83	0.09	.28	0.04	0.03	0.89
Maximum	1.55	−27.46	0.76	.98	0.37	0.23	7.64
Panel 2							
Mean	1.00	−29.04	0.18	.85	0.25	0.09	1.33
Median	1.01	−29.03	0.16	.90	0.23	0.08	1.30
Minimum	−0.18	−29.85	0.01	.05	0.02	0.01	0.11
Maximum	1.86	−28.50	0.54	1.00	0.77	0.27	3.18

[a] r^2 = coefficient of determination for regression of C13 ratio vs the environmental index for individual accessions.
[b] SE = standard error.
[c] −Log10(p) is the negative log (base 10) of the probability of the regressions of C13 ratio versus the environmental index for individual accessions. Shown are the mean, median, minimum, and maximum −Log10(p) values for the regressions of all 205 genotypes for Panel 1, and all 373 genotypes for Panel 2.

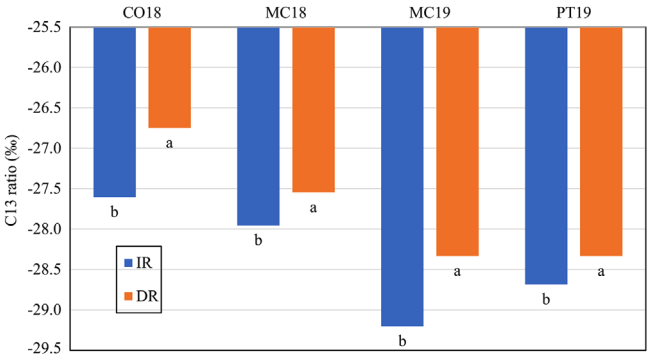


FIGURE 1 Average carbon isotope (C13) ratio of irrigated (IR) and drought (DR) treatments at four site-years of Panel 1 where both irrigated and drought treatments were present. The letters below each bar represent significant differences ($P < .01$) within an environment as determined from a T-test. CO18, Bradford Research Center at Columbia, MO in 2018; MC18, Maricopa Agricultural Center, University of Mariposa, AZ in 2018; MC19, Maricopa Agricultural Center, University of Mariposa, AZ in 2019; PT19, Pine Tree Research Station, AR in 2019

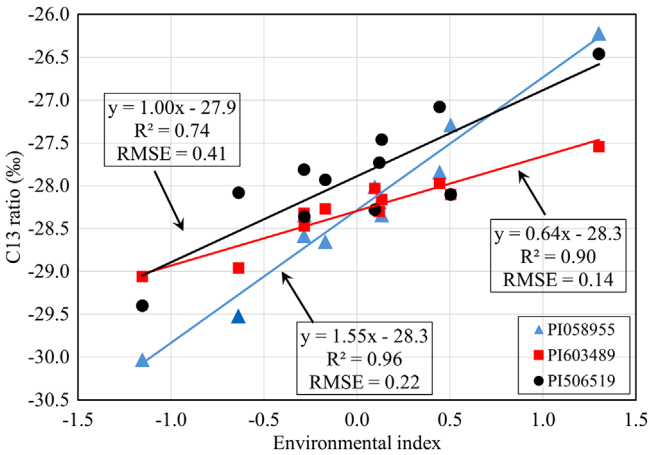


FIGURE 2 Selected responses of carbon isotope (C13) ratio versus the environmental index for three genotypes across 11 environments from Panel 1. The coefficient of determination (R^2) and the root mean square error (RMSE) for the regression equations are also presented

(Condon et al., 2002, 2004; Farquhar et al., 1982; Farquhar & Richards, 1984).

The ranges for plasticity, C13 ratio, and stability from the regression of C13 ratio vs EI across the 11 environments for individual accessions were 1.10, 1.37‰, and 0.67, respectively (Table 1; Supplemental Table S3; Supplemental Figure S1). Among the 205 genotypes evaluated, the regression of C13 ratio versus EI was significant ($P \leq .05$) for 200 genotypes. The proportion of variance (r^2) explained by individual regressions varied between .28 and .98, with 85% of the genotypes having an $r^2 \geq .70$ (Table 1; Supplemental Table S3).

Figure 2 shows the response of the C13 ratio versus EI for three genotypes from Panel 1 differing in their response across 11 environments. The slopes of the regression equations can be considered quantitative measures of phenotypic plasticity. The genotype represented by the red squares (PI603489), was relatively nonresponsive (less plastic) for C13 ratio across these 11 different environments with a slope of 0.64. In contrast, the genotype represented by the blue triangles (PI058955), had a slope of 1.55, meaning it was very plastic and highly responsive for C13 ratio across these

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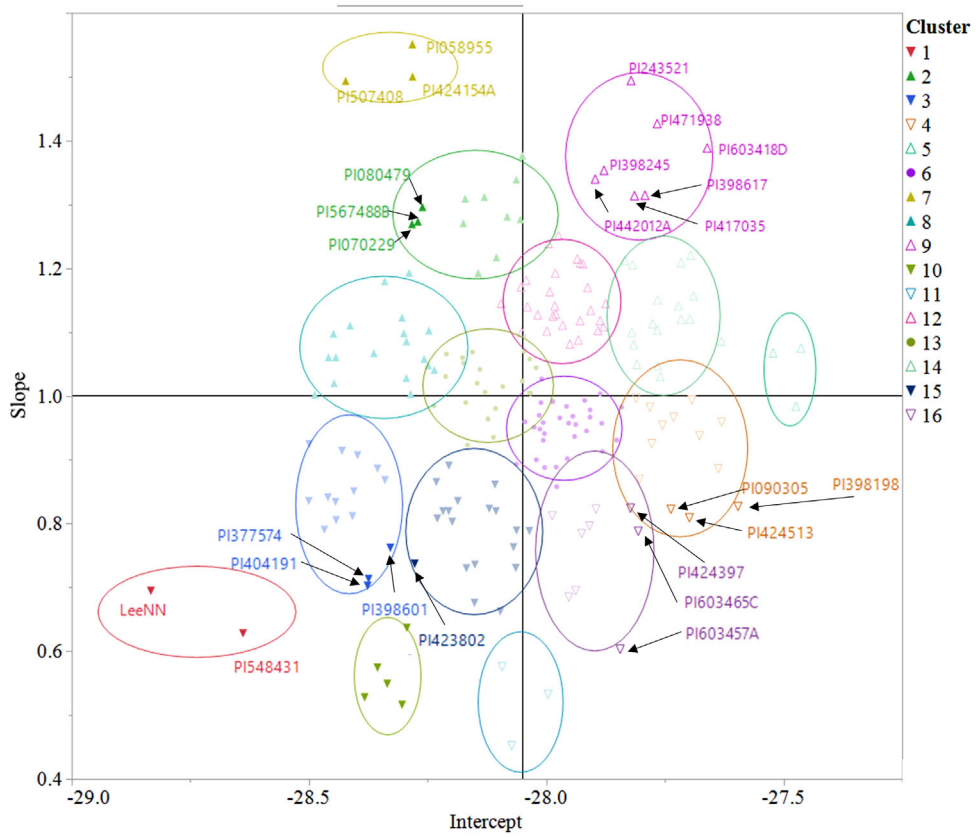


FIGURE 3 Biplot of the plasticity (slope) versus the carbon isotope (C13) ratio (intercept) from the regression of C13 ratio versus the environmental index for 11 environments for Panel 1. The 16 clusters were formed based on k-means clustering, and representative genotypes are shown in each quadrant. The horizontal and vertical lines indicate the average plasticity (slope) and C13 ratio (intercept) among the 205 genotypes across 11 environments

11 environments. The genotype represented by the black circles (PI506519), was intermediate in plasticity response indicated by a slope of 1.00. PI506519 had a greater intercept (−27.9‰) than both PI058955 (−28.3‰) and (PI603489 (−28.3‰), indicating a greater C13 ratio when considered over all environments. Although PI603489 was the least plastic genotype in Figure 2, it was also the most stable with a RMSE of 0.14 compared with a RMSE of 0.22 (PI058955) and 0.41 (PI506519). Results of the average C13 ratio and EI values for each environment along with the regressions for individual genotypes from Panels One and Two are provided in Supplemental Tables S2, S3, S4 and S5 respectively.

To identify genotypes with high plasticity for C13 ratio (large slope values from the regressions of C13 ratio vs the EI) that also had generally high C13 ratio (intercept) values, we created biplots of plasticity (slope values) versus C13 ratio (intercept) values (Figure 3) and used k-means clustering (Lloyd, 1982) to group genotypes with similar plasticity × C13 ratio responses. The resulting cluster outputs are presented in Supplemental Figure S2, and the quadrant where each genotype was found along with the k-means cluster number is indicated in Supplemental Table S3. Quadrant I of the biplot (Figure 3, upper right) indicates genotypes which feature both high C13 plasticity (steep slope) and the least negative C13 ratios (Clusters 5, 9, 12, and 14). Genotypes which cluster within Quadrant I are expected to have the most plastic responses to environment as well as the highest C13 ratio (i.e., WUE). Quadrant II (upper left) genotypes (Clusters 2, 7, and 8) are expected to be highly plastic but have relatively low C13 ratio (i.e., WUE). Genotypes in Quadrant III (bottom left, Clusters 1, 3, 10, and 15) are expected to be the least plastic and have low C13 ratio. Finally, genotypes that cluster in Quadrant IV (bottom right, Clusters 4 and 16) are predicted to have high C13 ratio but be relatively nonplastic in C13 ratio responses to different environments. Supplemental Table S3 identifies the quadrant and k-means cluster number for all genotypes in Panel 1. Genotypes of particular interest for future studies within each quadrant are identified in Figure 3 and described in more detail in Table 2.

3.1.2 | Panel 2

Environmental conditions for experiments conducted for Panel 2, including temperature, rainfall, and solar radiation, were described previously (Dhanapal et al., 2015a;

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TABLE 2 Representative genotypes for plasticity (slope) and carbon isotope (C13) ratio (intercept) values from the response of C13 ratio versus the environmental index over 11 environments from Panel 1. Shown are genotypes from each quadrant of the cluster analysis of the plasticity (slope) versus the C13 ratio (intercept)

Genotypes	Nobs used[a]	Plasticity	C13 ratio	Stability	r^2[b]	−Log10(p)	k-means cluster no.	Distance[c]
			‰					
Upper right-quadrant I (genotypes with high plasticity and high C13 ratio)								
PI398617	11	1.31	−27.79	0.24	.93	5.83	9	0.10
PI417035	11	1.31	−27.81	0.32	.89	4.82	9	0.10
PI603418D	11	1.39	−27.66	0.32	.90	4.99	9	0.37
PI243521	11	1.49	−27.82	0.21	.96	6.84	9	0.36
PI398245	11	1.35	−27.88	0.34	.88	4.64	9	0.11
PI471938	11	1.43	−27.77	0.26	.93	5.80	9	0.09
Upper left-quadrant II (genotypes with high plasticity and low C13 ratio)								
PI424154A	11	1.50	−28.28	0.49	.81	3.83	7	0.05
PI507408	10	1.49	−28.42	0.14	.97	6.78	7	0.17
PI567488B	11	1.27	−28.27	0.22	.94	5.97	2	0.27
PI058955	11	1.55	−28.28	0.22	.96	6.72	7	0.07
PI070229	11	1.27	−28.28	0.30	.89	4.93	2	0.33
PI080479	10	1.30	−28.26	0.38	.86	3.91	2	0.23
Lower left-quadrant III (genotypes with low plasticity and low C13 ratio)								
PI377574	11	0.76	−28.33	0.35	.68	2.74	3	0.25
PI404191	11	0.71	−28.37	0.41	.58	2.17	3	0.37
PI423802	11	0.74	−28.28	0.32	.71	2.95	15	0.35
PI548431	11	0.63	−28.64	0.35	.60	2.27	1	0.20
PI398601	9	0.70	−28.38	0.43	.42	1.24	3	0.43
LeeNN	11	0.70	−28.83	0.76	.28	1.02	1	0.20
Lower right-quadrant IV (genotypes with low plasticity and high C13 ratio)								
PI398198	10	0.83	−27.60	0.28	.82	3.52	4	0.47
PI424397	11	0.82	−27.82	0.19	.90	5.02	16	0.22
PI424513	11	0.81	−27.70	0.37	.69	2.78	4	0.30
PI603457A	11	0.60	−27.84	0.26	.71	2.97	16	0.64
PI090305	11	0.82	−27.74	0.40	.66	2.62	4	0.23
PI603465C	11	0.79	−27.81	0.42	.62	2.38	16	0.18

[a]Nobs used = number of observations used.
[b]r^2 = coefficient of determination for regression of C13 ratio vs the environmental index for individual accessions.
[c]Distance = squared Euclidean distance between the given observation and its cluster mean.

Kaler et al., 2017b). The C13 ratio averaged over genotypes within an environment ranged from −28.56‰ (ST09) to −29.56‰ (ST10; Supplemental TableS4). For the regression of C13 ratio versus EI for Panel 2, broad ranges of plasticity (2.04), C13 ratio (1.35‰), and stability (0.53) were observed (Table 1; Supplemental Figure S3). Of the 373 accessions, regressions for 268 accessions were significant ($P \leq .10$), but the coefficient of determination (r^2) was > .70 for 84% of the accessions.

Panel 2 genotypes with similar plasticity and C13 ratio were grouped as described for Panel 1 using cluster analysis (Figure 4 and Supplemental Figure S4). A subset of genotypes from each quadrant in Figure 4 was identified and described in more detail in Table 3. Supplemental TableS5 provides regression information of C13 ratio versus EI for each genotype in Panel 2 along with the k-means cluster number and the quadrant where each genotype was located from the biplot (Figure 4).

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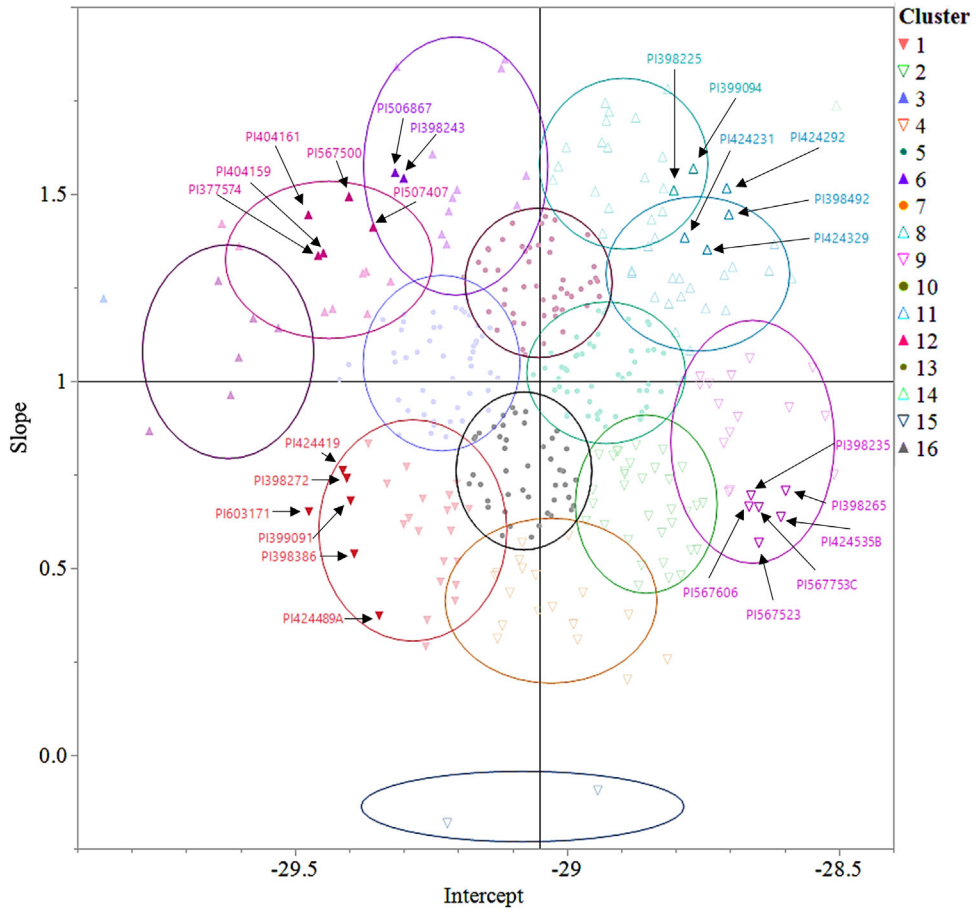


FIGURE 4 Biplot of the plasticity (slope) versus the carbon isotope (C13) ratio (intercept) from the regression of C13 ratio versus the environmental index for four environments for Panel 2. The 16 clusters were formed based on k-means clustering, and representative genotypes are shown in each quadrant. The horizontal and vertical lines indicate the average plasticity (slope) and C13 ratio (intercept) among the 373 genotypes across four environments

3.1.3 | Genome-wide association mapping for C13 plasticity, C13 ratio, and stability

In Panel 1, GWAM identified 45 significant SNPs at the threshold *P*-value [−Log10 (*P*) ≥ 3.5; *P* ≤ .0003]. Among these SNPs, there was one locus for plasticity (represented by 13 SNPs), 13 loci for C13 ratio (represented by 24 SNPs), and six loci for stability (represented by eight SNPs; Table 4; Supplemental Figure S5). The allelic effect of individual SNPs ranged from −0.15 to −0.11 for plasticity, −0.19 to 0.26‰ for C13 ratio, and −0.08 to 0.05 for stability.

In Panel 2, GWAM identified 29 SNPs at the threshold *P*-value [−Log10 (*P*) ≥ 3.5; *P* ≤ .0003]. There were six loci associated with plasticity (represented by six SNPs), 15 loci for C13 ratio (represented by 15 SNPs), and two loci for stability (represented by eight SNPs; Table 5; Supplemental Figure S6). The allelic effect of individual SNPs ranged from −0.16 to 0.28 for plasticity, from −0.04 to 0.04 for stability, and from −0.14 to 0.24‰ for C13 ratio.

3.1.4 | Candidate gene identification

To identify potential candidate genes, we examined the Glyma.Wm82.a2 version of soybean genome assembly (Schmutz et al., 2010) and examined a window size between ± 10 to 100 kb around each SNP. Candidate genes were identified, based on the presence of Gene Ontology (GO) terms annotated with direct or indirect association with drought and its related responses (root development and root hair elongation, auxin efflux transport, stomatal complex morphogenesis, leaf senescence and leaf vascular tissue formation, anthocyanin accumulation, response to heat, cold and salt, abscisic acid, jasmonic acid, auxin and gibberellic acid mediated signaling pathways). This approach identified 45 candidate genes from Panel 1 and 26 candidate genes from Panel 2 associated with either loci for plasticity, C13 ratio, or stability. Among these candidate genes, 17 and 15 SNPs were present within genes from Panel 1 and Panel 2, respectively. The annotations of the biological process, molecular

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TABLE 3 Representative genotypes for plasticity (slope) and carbon isotope (C13) ratio (intercept) values from the response of C13 ratio versus the environmental index over four environments from Panel 2. Shown are genotypes from each quadrant of the cluster analysis of the plasticity (slope) versus the C13 ratio (intercept)

Genotypes	Nobs used[a]	Plasticity	C13 ratio	Stability	r^2[b]	−Log10(p)	k-means cluster #	Distance[c]
			‰					
Upper right-quadrant I (genotypes with high plasticity and high C13 ratio)								
PI399094	4	1.57	−28.77	0.23	.92	1.38	8	0.35
PI398225	4	1.51	−28.80	0.16	.96	1.65	8	0.22
PI424231	4	1.38	−28.78	0.08	.99	2.16	11	0.09
PI398492	4	1.44	−28.70	0.16	.95	1.60	11	0.29
PI424292	4	1.52	−28.71	0.12	.97	1.89	11	0.52
PI424329	4	1.35	−28.74	0.15	.96	1.64	11	0.04
Upper left-quadrant II (genotypes with high plasticity and low C13 ratio)								
PI404161	4	1.44	−29.48	0.03	1.00	2.99	12	0.16
PI507407	4	1.41	−29.36	0.14	.96	1.73	12	0.21
PI506867	4	1.56	−29.32	0.13	.97	1.85	6	0.26
PI377574	4	1.34	−29.46	0.03	1.00	3.10	12	0.01
PI404159	4	1.34	−29.45	0.21	.91	1.33	12	0.01
PI567500	4	1.49	−29.40	0.10	.98	2.01	12	0.28
Lower left-quadrant III (genotypes with low plasticity and low C13 ratio)								
PI398386	4	0.54	−29.39	0.34	.39	0.43	1	0.28
PI398272	4	0.74	−29.40	0.23	.72	0.83	1	0.48
PI424489A	4	0.37	−29.35	0.18	.52	0.55	1	0.53
PI603171	4	0.65	−29.47	0.11	.90	1.31	1	0.79
PI399091	4	0.68	−29.40	0.12	.88	1.22	1	0.33
PI424419	4	0.76	−29.41	0.22	.75	0.88	1	0.57
Lower right-quadrant IV (genotypes with low plasticity and high C13 ratio)								
PI424535B	4	0.64	−28.61	0.03	.99	2.35	9	0.41
PI567753C	4	0.66	−28.65	0.23	.67	0.74	9	0.27
PI398235	4	0.69	−28.66	0.18	.79	0.96	9	0.18
PI567523	4	0.57	−28.65	0.02	.99	2.42	9	0.64
PI398265	4	0.71	−28.60	0.11	.91	1.34	9	0.23
PI567606	4	0.67	−28.67	0.14	.85	1.10	9	0.26

[a]Nobs used = number of observations used.
[b]r^2 = coefficient of determination for regression of C13 ratio vs the environmental index for individual accessions.
[c]Distance = squared Euclidean distance between the given observation and its cluster mean.

functions, and cellular components of these genes are reported in Supplemental Table S5 for Panel 1 and Supplemental Table S7 for Panel 2.

4 | DISCUSSION

The relationship between WUE and C13 ratio (or C13 isotopic discrimination) has provided an indirect selection tool for WUE that can be used for relatively large populations, but this comes with several caveats (Condon et al., 2004). The basis for the relationship between WUE and C13 ratio is that a decreased CO_2 concentration inside the leaf (C_i) results in an increased C13 ratio or decreased C13 discrimination (Farquhar & Richards, 1984; Farquhar et al., 1982). The application of this seemingly simple relationship to improve drought tolerance is complicated by several factors. A decrease in C_i may result from (a) partial stomatal closure, which results in relatively low photosynthetic activity, (b) highly efficient photosynthesis that decreases C_i despite having relatively high stomatal conductance, or (c) some combination of decreased stomatal conductance and high photosynthetic activity (Condon et al., 2004). These three scenarios that affect WUE may or may not be advantageous,

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TABLE 4 Significant single nucleotide polymorphisms (SNPs) associated with plasticity (slope), carbon isotope (C13) ratio (intercept), and stability (root mean square error, RMSE) from regression analysis of C13 ratio versus the environmental index for 11 environments from Panel 1 at the significant threshold *P*-value (−Log10 (*P*) ≥ 3.5; *P* ≤ .0003)

Locus	SNP_ID	CHR	Position	−Log10(P)	Allele[a]	Allelic effect[b]	Coincident genomic regions with previous studies[c]
						‰	
Plasticity							
1	ss715614110	13	21,632,895	4.70	C/T	−0.15	
	ss715614057	13	21,648,351	4.52	G/A	−0.14	
	ss715614053	13	21,649,205	4.09	G/A	−0.13	
	ss715614000	13	21,665,665	4.04	C/T	−0.13	
	ss715613955	13	21,683,574	3.62	T/C	−0.14	
	ss715616216	13	21,821,538	3.86	A/C	−0.14	
	ss715616141	13	21,822,239	3.86	C/T	−0.14	
	ss715615946	13	21,824,541	3.86	T/C	−0.14	
	ss715615841	13	21,825,102	3.86	A/C	−0.14	
	ss715615791	13	21,825,514	3.86	A/C	−0.14	
	ss715614949	13	21,831,889	4.52	A/C	−0.13	
	ss715613868	13	21,845,683	4.05	T/C	−0.12	
	ss715616781	13	21,855,479	3.98	A/G	−0.11	
C13 ratio							
1	ss715579421	1	43,240,525	4.63	C/T	−0.19	GWAS-CW QTL[7]
2	ss715583236	2	45,780,363	3.83	C/A	−0.13	GWAS-CW QTL[2]
	ss715583237	2	45,785,740	3.79	T/C	−0.12	GWAS-CW QTL[2]
	ss715583238	2	45,789,843	3.83	T/G	−0.13	GWAS-CW QTL[2]
	ss715583239	2	45,795,780	3.79	A/G	−0.12	GWAS-CW QTL[2]
	ss715583240	2	45,807,464	3.54	T/C	−0.13	GWAS-CW QTL[2]
	ss715583242	2	45,820,474	3.64	A/G	−0.14	GWAS-CW QTL[2]
	ss715583243	2	45,824,097	3.64	C/T	−0.14	GWAS-CW QTL[2]
	ss715583244	2	45,832,608	3.64	G/A	−0.14	GWAS-CW QTL[2]
	ss715583245	2	45,833,086	3.61	C/T	−0.12	GWAS-CW QTL[2]
	ss715583246	2	45,837,555	3.61	C/T	−0.12	GWAS-CW QTL[2]
3	ss715586494	3	44,251,912	3.84	T/C	−0.06	
	ss715586496	3	44,260,819	3.84	T/G	−0.06	
	ss715586497	3	44,264,670	3.84	T/C	−0.06	

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TABLE 4 (Continued)

Locus	SNP_ID	CHR	Position	−Log10(P)	Allele[a]	Allelic effect[b]	Coincident genomic regions with previous studies[c]
4	ss715594630	6	47,112,305	5.9	A/C	0.07	
5	ss715602966	9	1,124,625	8.11	C/T	0.26	GWAS-CW QTL[2,7], GWAS-C13 & O18 QTL[8]
6	ss715606980	10	40,649,350	5.23	T/C	−0.06	
7	ss715612687	12	36,530,315	4.29	C/T	0.22	
8	ss715615512	13	34,363,427	5.52	T/G	−0.05	
9	ss715619888	14	758,810	3.91	C/A	0.26	GWAS-CW QTL[2]
10	ss715622332	15	4,770,814	3.76	G/A	0.04	GWAS-CW QTL[7]
11	ss715623306	15	9,983,279	6.49	G/A	0.24	
12	ss715635425	19	45,204,441	7.63	C/A	0.26	GWAS-CW QTL[2, 3]
13	ss715638759	20	46,499,240	3.55	G/A	−0.02	GWAS-CW QTL[3]
Stability							
1	ss715594151	6	3,226,750	3.83	G/A	0.05	
2	ss715597694	7	38,021,435	4.32	T/G	−0.08	
	ss715597719	7	38,132,939	3.54	A/G	−0.07	
	ss715597730	7	38,181,668	3.54	C/T	−0.07	
3	ss715597976	7	40,573,526	3.74	G/A	−0.05	
4	ss715605228	9	6,045,623	3.79	A/G	0.05	
5	ss715621140	15	17,846,447	4.34	A/G	−0.06	
6	ss715630350	18	3,527,395	3.69	C/T	-0.05	GWAS-CW QTL[7]

Note. For each SNP, the following information is reported: locus number, chromosome (CHR) and position in base pairs, allele and allelic effect, and overlapping quantitative trait loci (QTLs) for C13 and oxygen isotope (O18) ratios, canopy wilting (CW), and canopy temperature (CT) from previous studies. Individual SNPs were assigned to a locus if they were located in the same linkage disequilibrium block of ± 175 kb.

[a]Allele: major/minor alleles of SNP.

[b]Allelic effect: difference in mean C13 ratio composition between genotypes with major allele and minor allele. Positive sign indicates that the major allele is associated with increased C13 ratio. Negative sign indicates that the minor allele is associated with increased C13 ratio.

[c]Coincident genomic regions with previous studies: (1) SNPs from Panel 2 of current study (C13); (2) Chamarthi et al. (2021) (CW); (3) Steketee et al. (a2020) (CW); (4) Bazzer et al. (2020b) (C13); (5) Bazzer & Purcell (2020) (CT); (6) Kaler et al. (2018b) (CT); (7) Kaler et al. (2017a) (CW); (8) Kaler et al. (2017b) (C13 & O18). Coincident regions identified preceded by 'GWAS' were identified by association mapping; otherwise, QTLs were identified in biparental populations.

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TABLE 5 Significant single nucleotide polymorphisms (SNPs) associated with plasticity (slope), carbon isotope (C13) ratio (intercept), and stability (root mean square error, RMSE) from regression analysis of C13 ratio versus the environmental index for four environments from Panel 2 at the significant threshold *P*-value (−Log10 (*P*) ≥ 3.5; *P* ≤ .0003)

Locus	SNP_ID	CHR	Position	−Log10(P)	Allele[a]	Allelic effect[b]	Coincident genomic regions with previous studies[c]
Plasticity							
1	ss715582739	2	4,486,771	3.68	T/C	0.28	GWAS-CW QTL[2], GWAS-C13 QTL[8]
2	ss715588201	4	45,359,800	3.93	C/T	−0.08	
3	ss715595088	6	49,689,167	4.13	G/A	0.02	
4	ss715603233	9	1,943,831	7.37	G/A	0.13	GWAS-CW QTL[3], GWAS-C13 QTL[8]
5	ss715609266	11	25,290,358	4.74	T/C	0.08	
6	ss715613885	13	20,242,637	3.52	C/A	−0.16	GWAS-CW QTL[7]
C13 ratio							
1	ss715582962	2	43,112,945	4.32	C/T	0.24	GWAS-C13 QTL[8]
2	ss715589227	4	7,128,216	4.64	G/A	0.16	GWAS-CW QTL[2]
3	ss715588669	4	49,127,886	5.41	A/C	0.09	GWAS-CT QTL[6], GWAS-CW QTL[7], GWAS-C13 QTL[8]
4	ss715588863	4	50,475,329	3.98	G/T	−0.13	GWAS-C13 QTL[8]
5	ss715592198	5	38,868,670	7.05	G/A	−0.12	GWAS-C13 QTL[8]
6	ss715595870	7	120,262	3.59	A/G	0.07	
7	ss715607253	10	43,167,977	3.54	G/A	0.12	
8	ss715609311	11	1,626,832	6.17	C/T	0.09	GWAS-CW QTL[2]
9	ss715613325	12	6,772,514	4.60	T/C	0.15	
10	ss715613850	13	2,749,908	4.74	G/T	−0.14	
11	ss715615368	13	33,018,192	3.61	G/A	−0.13	
12	ss715624450	16	31,818,520	3.88	A/G	0.19	GWAS-C13 QTL[8]
13	ss715634299	19	35,676,831	3.93	A/G	0.01	
14	ss715637669	20	36,242,932	4.07	A/G	−0.10	
15	ss715638812	20	46,851,986	4.67	T/C	0.05	

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TABLE 5 (Continued)

Locus	SNP_ID	CHR	Position	−Log10(P)	Allele[a]	Allelic effect[b]	Coincident genomic regions with previous studies[c]
Stability							
1	ss715593249	6	14,533,063	3.59	A/G	0.04	GWAS-CW QTL[2, 3]
	ss715593254	6	14,547,249	3.59	A/G	0.04	GWAS-CW QTL[2, 3,] C13 QTL[4]
	ss715593256	6	14,554,440	3.69	C/T	0.04	GWAS-CW QTL[2, 3,] C13 QTL[4]
	ss715593257	6	14,554,693	3.59	A/G	0.04	GWAS-CW QTL[2, 3,] C13 QTL[4]
2	ss715613360	12	6,999,850	3.94	G/T	−0.04	
	ss715613362	12	7,008,714	3.94	G/T	−0.04	
	ss715613367	12	7,078,699	3.87	A/G	−0.04	
	ss715613370	12	7,082,204	3.87	G/A	−0.04	

Note. For each SNP, the following information is reported: locus number, chromosome (CHR) and position in base pairs, allele and allelic effect, and overlapping quantitative trait loci (QTLs) for C13 and oxygen isotope (O18) ratios, canopy wilting (CW), and canopy temperature (CT) from previous studies. Individual SNPs were assigned to a locus if they were located in the same linkage disequilibrium block of ± 175 kb.

[a]Allele: major/minor alleles of SNP.

[b]Allelic effect: difference in mean C13 ratio composition between genotypes with major allele and minor allele. Positive sign indicates that the major allele is associated with increased C13 ratio. Negative sign indicates that the minor allele is associated with increased C13 ratio.

[c]Coincident genomic regions with previous studies: (1) SNPs from Panel 1 of current study (C13); (2) Chamarthi et al. (2021) (CW); (3) Steketee et al. (2020) (CW); (4) Bazzer et al. (2020b) (C13); (5) Bazzer & Purcell (2020) (CT); (6) Kaler et al. (2018b) (CT); (7) Kaler et al. (2017a) (CW); (8) Kaler et al. (2017b) (C13 & O18). Coincident regions identified preceded by 'GWAS' were identified by association mapping; otherwise, QTLs were identified in biparental populations.

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depending upon the specific environment. Low stomatal conductance may conserve soil moisture and be an advantage in environments where soil moisture is limited, but this mechanism would likely limit photosynthesis and crop growth in water-replete environments. Therefore, selection for genotypes that constitutively have a low or a high C13 ratio would unlikely have wide adaptation across diverse environments. The main objective of the current research was to identify genotypes, and SNPs linked to favorable alleles associated with genotypic responsiveness for C13 ratio (i.e., highly plastic) across different environments, and ideally to provide high WUE in drought environments and low WUE under water-replete environments.

Phenotypic stability has been defined in several different ways (Bernardo, 2002; Eberhart & Russell, 1966). Type 1 stability considers the slope of the regression for a genotype versus EI with a value close to 0 indicating stability; Type 2 stability considers a genotype stable when the slope is similar to the average slope of all genotypes (i.e., slope = 1); a genotype with Type 3 stability has the variance or RMSE of the regression equation approaching 0. Here, we explicitly defined stability as the RMSE (Type 3) of the regression, plasticity as the slope of C13 ratio versus the EI, and the intercept as an average C13 ratio across all environments (C13 ratio). Frequency distributions of the plasticity, C13 ratio, and stability for genotypes in both Panel 1 and Panel 2 all had a wide range of continuous values (Supplemental Figures S1 and S3), indicating quantitative inheritance of complex traits that are suitable for association mapping (McCarthy et al., 2008).

Association mapping has been used to identify loci associated with plasticity in several crops. Kadam et al. (2017) used a diverse panel of 274 *Oryza indica* genotypes grown under irrigated and drought conditions and defined plasticity as the change in trait response between irrigated and drought treatments relative to the irrigated treatment; from this they identified 76 loci related to root morphology and 233 candidate genes associated with anatomical plasticity and root morphology. Mangin et al. (2017) combined crop modeling with phenotypic variation to the environment and identified nine loci associated with oil yield plasticity in sunflower hybrids under cold stress using the slopes from a Finlay and Wilkinson (1963) analysis as a measure of plasticity. Kusmec et al. (2017) evaluated phenotypic plasticity for several traits in a maize nested association mapping (NAM) population under multiple environments using a Bayesian Finlay–Wilkinson regression procedure. Their results indicated that the candidate genes for mean phenotypic values and plasticity measures were distinct groups, both structurally and functionally.

Previous research in soybean has identified QTLs associated with C13 ratio (or C13 discrimination) using association mapping (Dhanapal et al., 2015a; Kaler et al., 2017b) and biparental mapping (Bazzer et al., 2020a, 2020b) that were identified from specific environments or averaged across several environments. For both the association mapping and biparental mapping reports of C13 ratio, broad- and narrow-sense heritabilities across environments were between 76 and 90%, but in all cases effects of G×E interactions were significant ($P \leq .0001$). Consistent with the G×E interactions for individual genotypes is the wide range of plasticity and C13 ratio values among genotypes in the current study, and that 200 of 205 genotypes from Panel 1 had significant ($P \leq .05$) regressions (Supplemental Table S3) and that 72% of the genotypes from Panel 2 had significant ($P \leq .10$) regressions (Supplemental Table S5).

The genotypes highlighted in the biplot of the slope value (i.e., plasticity) versus the C13 ratio (i.e., intercept) values of the regression analysis for Panel 1 and Panel 2 (Figures 3 and 4; Tables 2 and 3) were compared with extremes for canopy wilting reported previously from association mapping that included many of the same genotypes used in the present study (Chamarthi et al., 2021; Kaler et al., 2017a). Because genotypes in Quadrants I and IV of the biplots had relatively high C13 ratios (Figures 3 and 4), they were predicted to have high WUE, conserve soil moisture during drought, and be slow wilting, whereas genotypes in Quadrants II and III had relatively low values of C13 ratio and were predicted to have low WUE, rapidly deplete soil moisture, and wilt quickly. PI507408 and PI398601 from Panel 1 were in Quadrants II and III (Figure 3; Table 2), respectively, and as predicted were previously reported as fast-wilting extremes (Chamarthi et al., 2021). Also, in Panel 1, PI603457A in Quadrant IV was previously identified as a slow-wilting extreme (Chamarthi et al., 2021). Similarly, for Panel 2, PI404159 and PI506867 were in Quadrant II (Figure 4; Table 3) and, as predicted, were previously identified as fast-wilting extremes (Kaler et al., 2017a). Furthermore, PI507407 and PI567753C in Quadrants II and IV were identified previously as extremes for fast and slow wilting, respectively (Chamarthi et al., 2021; Kaler et al., 2017a).

Slow-wilting check genotypes from Panel 1, PI416937 and PI471938, had among the greatest values of plasticity (slopes of 1.34 and 1.43, respectively), whereas the intercept values were −28.06 and −27.77‰, respectively. The large plasticity of these two genotypes would allow increased C13 ratio (i.e., increased WUE) under drought conditions and decreased C13 ratio when soil moisture was not limiting. In contrast, slope values for the two fast-wilting checks 08705_16 and A5959 were 1.10 and 1.17, respectively, and intercept values were −28.45 and −28.05‰. Hence both fast-wilting checks were located in Quadrant II and projected to be fast wilting.

Association mapping of plasticity, C13 ratio, and stability identified one, 13, and six loci based on Panel 1, and six, 15, and two loci based on Panel 2, respectively. Despite several of the loci from Panels One and Two for plasticity, C13 ratio, and stability overlapping with loci identified previously

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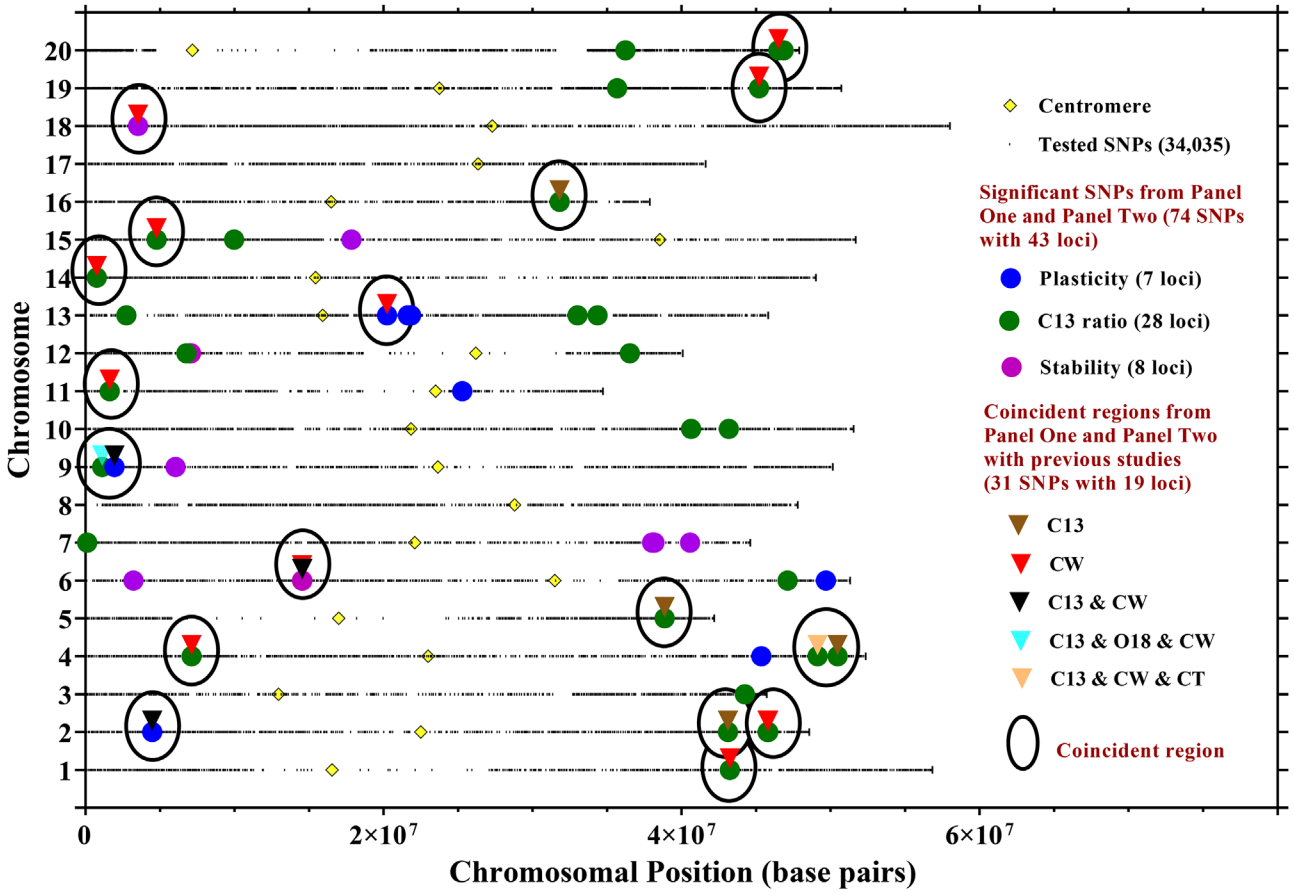


FIGURE 5 Location of single nucleotide polymorphisms (SNPs) significantly associated with plasticity (slope), carbon isotope (C13) ratio (intercept), and stability (root mean square error, RMSE) values of Panel 1 and Panel 2, and coincident genomic regions from Panel 1 and Panel 2 with previous studies. Coincident genomic regions from previous studies for C13 ratio (C13) are shown from Bazzer et al. (2020a, 2020b) and Kaler et al. (2017b); coincident regions for canopy temperature (CT) are shown from Bazzer & Purcell (2020) and Kaler et al. (2018b); coincident regions for canopy wilting (CW) are shown for Chamarthi et al. (2021), Kaler et al. (2017a), and Steketee et al. (2020); coincident regions for both C13 ratio and oxygen isotope (O18) ratio (C13 & O18) are shown for Kaler et al. (2017b)

(Tables 4 and 5), none of the loci between Panels One and Two overlapped. It is uncertain why no common loci were found for the two panels, but differences between environments and the range of conditions encompassed by the environments, as illustrated by the range in temperature and precipitation as well as EIs (Supplemental Tables S1, S2, and S4) for these panels likely contributed to the absence of coincident loci.

Association mapping of plasticity (i.e., slope values) identified one locus from Panel 1 and six loci from Panel 2 (Tables 4 and 5) that cumulatively had relatively large effects. There were 13 significant SNPs in one locus on Gm13 from Panel 1 that had an allelic effect that averaged 0.13. The total range of plasticity values from Panel 1 was 1.10, indicating that the allelic effect of the single locus on Gm13 accounted for about 12% of the genotypic differences observed. Likewise, for Panel 2, the six loci associated with plasticity had a cumulative allelic effect of 0.75, which accounted for approximately 37% of the range of plasticity values (2.04) from Panel 2. It is noteworthy that the locus on Gm02 from Panel 2 had a particularly large allelic effect of 0.28. Taking the same approach for the C13 ratio, the cumulative allelic effect of loci for the C13 ratio (i.e., intercept) of Panel 1 and Panel 2 accounted for 136 and 133% of the range of values observed, respectively, indicating that no one genotype had all the best or worst alleles for these traits. For stability (i.e., RMSE), the cumulative allelic effect of loci accounted for 49 and 15% of the range for Panels 1 and 2, respectively.

Among the 45 candidate genes identified from Panel 1, several were of particular interest. Associated with plasticity on Gm13 (ss715613955) was *Glyma.13g101900* which belongs to the auxin efflux carrier family protein (Supplemental Table S6). Auxin is a phytohormone that plays key roles in plant growth and development, and polar auxin transport is involved in response to drought in plants. Its distribution largely depends on the polar localization of the PIN-FORMED (PIN) auxin efflux carrier family of proteins. Over expression of the *OsPIN3t* gene leads to improved drought tolerance and drought stress response in rice

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(Zhang et al., 2012). Similarly, a SNP associated with C13 ratio intercept on Gm13 (ss715615512) was *Glyma.13g232500* which belongs to magnesium (Mg) chelatase activity (Supplemental Table S6). The H subunit of Mg-chelatase (*CHLH*) regulates abscisic acid (ABA) signaling and the I subunit (*CHLI*) modulates ABA signaling in guard cells. A new rtl1 mutant allele of the *CHLH* gene in *Arabidopsis thaliana* showed ABA insensitive phenotypes in both stomatal movement and seed germination. Upregulation of *CHLI1* resulted in ABA hypersensitivity in seed germination, whereas downregulation of *CHLI* conferred ABA insensitivity in stomatal response in Arabidopsis. Further, *CHLH* and *CHLI* also regulate stomatal sensitivity to ABA in tobacco (*Nicotiana benthamiana*; Du et al., 2012). Additionally, a SNP associated with C13 ratio intercept on Gm19 (ss715635425) was *Glyma 19g194500* which belongs to the basic Leucine Zipper (bZIP) transcription factor family protein (Supplemental Table S6). The ABA signaling pathway consists of receptors, phosphatases, kinases, and transcription factors, among them ABA Insensitive 5 (ABI5) and Abre Binding Factors/Abre-Binding Proteins (ABFs/AREBs), which belong to the bZIP family and which control expression of stress-responsive genes. ABI5 is mostly active in seeds and prevents germination and post-germinative growth under unfavorable conditions. The ABFs/AREBs act redundantly in regulating genes that control physiological processes in response to stress during vegetative growth. Overexpression of genes encoding ABA-dependent bZIPs increase stress tolerance of transgenic plants (Collin et al., 2021).

Of the 43 loci associated with plasticity, C13 ratio, and stability for Panels 1 and 2, 19 loci were within the same LD block as QTLs reported previously for either C13 ratio (Bazzer et al., 2020b; Kaler et al., 2017b), canopy wilting (Chamarthi et al., 2021; Kaler et al., 2017a; Steketee et al., 2020), canopy temperature (Bazzer & Purcell, 2020; Kaler et al., 2018b), or both C13 ratio and O18 ratio (Kaler et al., 2017b; Tables 4 and 5; Figure 5). A high C13 ratio would expectedly result from partial stomatal closure (Farquhar & Richards, 1984; Farquhar & Sharkey, 1982; Farquhar et al., 1989), decreasing transpiration and conserving soil moisture. Previous research has characterized slow-wilting genotypes as having greater soil moisture (King et al., 2009; Ries et al., 2012), greater C13 ratio (Bai & Purcell, 2018a), and cooler canopy temperatures (Bai & Purcell, 2018b) under drought stress than fast-wilting genotypes; as such it is not surprising that QTLs for C13 ratio would have pleiotropic effects on canopy wilting and canopy temperature.

In summary, the ratio of ^{13}C to ^{12}C (C13 ratio) has been closely associated with WUE in soybean and other C3 plants, but its practical application for improving crop performance under drought has been limited in part because increased WUE likely restricts photosynthesis and yield when soil moisture is not limiting. Our goal was to identify genotypes (and the responsible alleles) that respond to drought environments by increasing C13 ratio (WUE), while maintaining high photosynthetic rates (i.e., low C13 ratio and low WUE) when soil moisture was not limiting. We characterized plasticity of C13 ratio for two panels of MG IV soybean genotypes using an Eberhart–Russell regression analysis of C13 ratio versus an EI. This analysis identified genotypes with a wide range of plasticity (slope), average C13 ratio over environments (intercept), and stability (RMSE). Using the slope, intercept, and RMSE values from the regression analysis of each genotype, we identified loci (SNPs) associated with plasticity (7 loci), C13 ratio (28 loci), and stability (8 loci). These results provide the necessary information to utilize high WUE in soybean using C13 ratio without compromising yield potential in water-replete environments, as well as identifying genotypes varying in C13 ratio plasticity and stability, and the underlying loci for these traits.

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AUTHOR CONTRIBUTIONS

Avjinder S. Kaler: Data curation; Investigation; Writing – review & editing. Hussein Abdel-Haleem: Data curation; Investigation; Writing – review & editing. Felix B. Fritschi: Conceptualization; Data curation; Investigation; Validation; Visualization; Writing – review & editing. Jason D. Gillman: Data curation; Formal analysis; Investigation; Validation; Visualization; Writing – review & editing. Jeffery D. Ray: Data curation; Investigation; Validation; Visualization; Writing – review & editing. Larry C. Purcell: Data curation; Formal analysis; Funding acquisition; Investigation; Methodology; Project administration; Supervision; Writing – original draft; Writing – review & editing.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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